SQL Command Categories

SQL commands are divided into **5 categories**.

Category	Full Form	Purpose
DDL	Data Definition Language	Creates and modifies database structure
DML	Data Manipulation Language	Inserts, updates, and deletes data
DQL	Data Query Language	Retrieves data
DCL	Data Control Language	Controls user permissions
TCL	Transaction Control Language	Controls transactions

---

1. DDL (Data Definition Language)

DDL commands work on the **structure** of the database.

Think of DDL as **building the house**, not putting furniture inside.

Commands:

- CREATE
- ALTER
- DROP
- TRUNCATE
- RENAME

CREATE

Used to create

- Database
- Table
- View
- Index

Create Database

```
CREATE DATABASE college;
```

Use Database

```
USE college;
```

Create Table

```
CREATE TABLE students (
    id INT,
    name VARCHAR(50),
    age INT
);
```

Output

id name age

Table created successfully.

Data Types

Numeric

```
INT
BIGINT
FLOAT
DOUBLE
DECIMAL(10,2)
```

Example

```
salary DECIMAL(10,2)
```

Stores

```
25000.75
150000.50
```

Character

```
CHAR(10)
VARCHAR(100)
TEXT
```

Difference

CHAR	VARCHAR
Fixed length	Variable length
Faster	Saves space

Example

```
name VARCHAR(100)
```

Date & Time

```
DATE
TIME
DATETIME
TIMESTAMP
YEAR
```

Example

```
joining_date DATE
```

Constraints

Constraints are rules applied on columns.

Common Constraints

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE
- DEFAULT
- CHECK
- AUTO_INCREMENT

PRIMARY KEY

Rules

- Unique
- Cannot be NULL
- Only one primary key per table

Example

```
CREATE TABLE students(
    id INT PRIMARY KEY,
    name VARCHAR(50)
);
```

Valid

```
1
2
3
```

Invalid

```
1
1
NULL
```

AUTO_INCREMENT

Automatically increases value.

```
CREATE TABLE students(
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50)
);
```

Insert

```
INSERT INTO students(name)
VALUES("Akshat");
```

Output

```
id = 1
```

Next insert

```
id = 2
```

NOT NULL

Column cannot be empty.

```
CREATE TABLE students(
    id INT,
    name VARCHAR(50) NOT NULL
);
```

Wrong

```
INSERT INTO students
VALUES(1,NULL);
```

Error

UNIQUE

Duplicate values are not allowed.

```
email VARCHAR(100) UNIQUE
```

Valid

```
abc@gmail.com
xyz@gmail.com
```

Invalid

```
abc@gmail.com
abc@gmail.com
```

DEFAULT

Assigns default value.

```
status VARCHAR(20)
DEFAULT 'Active'
```

Insert

```
INSERT INTO students(name)
VALUES("Rahul");
```

Result

```
Status = Active
```

CHECK

Applies condition.

```
age INT CHECK(age>=18)
```

Allowed

```
18
20
25
```

Not Allowed

```
15
16
```

ALTER

Used to modify table.

Operations

- Add column
- Modify column
- Rename column
- Drop column

Add Column

```
ALTER TABLE students
ADD email VARCHAR(100);
```

Modify Column

```
ALTER TABLE students
MODIFY age BIGINT;
```

Rename Column

```
ALTER TABLE students
RENAME COLUMN name TO student_name;
```

Drop Column

```
ALTER TABLE students
DROP COLUMN email;
```

RENAME TABLE

```
RENAME TABLE students TO student_details;
```

TRUNCATE

Deletes all records.

Structure remains.

```
TRUNCATE TABLE students;
```

Before

```
5 rows
```

After

```
0 rows
```

Table still exists.

DROP

Deletes everything.

- Table
- Structure
- Data

```
DROP TABLE students;
```

Everything is deleted.

Difference

DELETE vs TRUNCATE vs DROP

DELETE	TRUNCATE	DROP
Deletes selected rows	Deletes all rows	Deletes whole table
WHERE allowed	WHERE not allowed	Removes structure
Can rollback	Usually cannot	Cannot rollback
Table remains	Table remains	Table removed

---

2. DML (Data Manipulation Language)

Works with data.

Commands

- INSERT
- UPDATE
- DELETE

INSERT

Adds records.

```
INSERT INTO students
VALUES(1,"Akshat",22);
```

Insert multiple rows

```
INSERT INTO students
VALUES
(2,"Rahul",21),
(3,"Neha",20),
(4,"Priya",22);
```

Insert using column names

```
INSERT INTO students(name,age)
VALUES("Amit",25);
```

UPDATE

Changes existing records.

```
UPDATE students
SET age=23
WHERE id=1;
```

Without WHERE

```
UPDATE students
SET age=20;
```

All rows become 20.

DELETE

Deletes records.

```
DELETE FROM students
WHERE id=2;
```

Delete all rows

```
DELETE FROM students;
```

Table remains.

---

3. DQL (Data Query Language)

Main command

```
SELECT
```

Retrieve all data

```
SELECT * FROM students;
```

Retrieve specific columns

```
SELECT name,age
FROM students;
```

WHERE

Filters records.

```
SELECT *
FROM students
WHERE age>20;
```

Operators

```
=
>
<
>=
<=
!=
<>
```

Logical Operators

```
AND
OR
NOT
```

Example

```
SELECT *
FROM students
WHERE age>18
AND city='Delhi';
```

ORDER BY

Ascending

```
SELECT *
FROM students
ORDER BY age ASC;
```

Descending

```
SELECT *
FROM students
ORDER BY age DESC;
```

LIMIT

```
SELECT *
FROM students
LIMIT 5;
```

Returns first five rows.

DISTINCT

Removes duplicate values.

```
SELECT DISTINCT city
FROM students;
```

---

4. DCL (Data Control Language)

Controls permissions.

Commands

```
GRANT
REVOKE
```

Grant permission

```
GRANT SELECT
ON students
TO user1;
```

Remove permission

```
REVOKE SELECT
ON students
FROM user1;
```

---

5. TCL (Transaction Control Language)

Used to manage transactions.

Commands

- COMMIT
- ROLLBACK

- SAVEPOINT

COMMIT

Permanently saves changes.

```
COMMIT;
```

ROLLBACK

Undo changes before commit.

```
ROLLBACK;
```

SAVEPOINT

Creates a checkpoint.

```
SAVEPOINT sp1;
```

Rollback to savepoint

```
ROLLBACK TO sp1;
```

Transaction Example

```
START TRANSACTION;

UPDATE accounts
SET balance = balance - 500
WHERE id = 1;

UPDATE accounts
SET balance = balance + 500
WHERE id = 2;

COMMIT;
```

If an error occurs

```
ROLLBACK;
```

Complete Flow

```
CREATE DATABASE college;

USE college;

CREATE TABLE students(
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100) NOT NULL,
email VARCHAR(100) UNIQUE,
age INT CHECK(age>=18),
city VARCHAR(50) DEFAULT 'Delhi'
```

```
);

INSERT INTO students(name,email,age)
VALUES
('Akshat','akshat@gmail.com',22),
('Rahul','rahul@gmail.com',21);

SELECT * FROM students;

UPDATE students
SET city='Rohtak'
WHERE id=1;

DELETE FROM students
WHERE id=2;

ALTER TABLE students
ADD phone VARCHAR(20);

SELECT * FROM students;
```

Interview Questions

Q1. What is the difference between DDL and DML?

DDL modifies the structure of database objects (tables, columns, databases), while **DML** works with the data stored inside those objects.

Q2. Difference between DELETE, TRUNCATE, and DROP?

- **DELETE:** Removes selected rows; `WHERE` clause can be used.
- **TRUNCATE:** Removes all rows quickly but keeps the table structure.
- **DROP:** Deletes both the table structure and all its data.

Q3. What is a Primary Key?

A **Primary Key** uniquely identifies each record in a table. It cannot contain duplicate or `NULL` values.

Q4. What is the difference between CHAR and VARCHAR?

- **CHAR:** Fixed-length string.
- **VARCHAR:** Variable-length string that uses only the required storage.

Q5. Why is the WHERE clause important in UPDATE and DELETE?

Without a `WHERE` clause, **all rows** in the table are updated or deleted, which can lead to accidental data loss.

Quick Revision

Command	Purpose

Command	Purpose
CREATE	Create database/table
ALTER	Modify table structure
DROP	Delete table completely
TRUNCATE	Remove all rows, keep structure
RENAME	Rename table
INSERT	Add records
UPDATE	Modify records
DELETE	Remove records
SELECT	Retrieve records
GRANT	Give permissions
REVOKE	Remove permissions
COMMIT	Save transaction
ROLLBACK	Undo transaction
SAVEPOINT	Create a rollback checkpoint